Hyun In Jo, Ph. D., LEED A.P., WELL A.P.

Senior Research Engineer, Noise-Vibration-Harshness (NVH) Test Research Team
Automotive, R&D Division, Hyundai Motor Company
150, Hyundaiyeonguso-ro, Namyang-eup, Hwaseong-si, Gyeonggi-do, 18280, Korea
Phone: +82-10-6387-8402, Email: best2012@naver.com, ORCID number: 0000-0002-4243-3352

EDUCATION

Ph.D. and M.S. Integrated Degree 2016.03. – 2022.02.	**Department of Architectural Engineering** Architectural Environmental System (Architectural acoustics / Soundscape) Hanyang University, Korea (GPA: 4.39 / 4.5) • Thesis: Psycho-physiological Restoration with Audio-visual Interaction in Urban Soundscape Experience • Advisor: Jin Yong Jeon
B.S. 2013.03. – 2016.02.	**Department of Architectural Engineering** Hanyang University, Korea (GPA: 4.23 / 4.5) • Graduation Summa Cum Laude (Early graduation)

ACADEMIC AND PROFESSIONAL WORK EXPERIENCES

2022.09. – Present	**Senior Research Engineer**, Noise-Vibration-Harshness (NVH) Research Lab *Hyundai Motor Group, Namyang R&D Center, Korea.*
2022.03. – 2022.08.	**Young Scientist Researcher (Post-doc)**, Department of Building Energy Research *The Korea Institute of Civil Engineering and Building Technology, Ilsan, Korea.*
2018.10. – 2018.10.	**Visiting Researcher**, Institut Jean Le Rond d'Alembert *Sorbonne University, Paris, France.*
2016.01. – 2022.02.	**Research Assistant**, Architectural Acoustics Lab *Hanyang University, Korea*
2015.06. – 2015.08.	**Research Assistant**, Internship *Architectural Acoustic Research Center, Seoul, Korea.*

INTERNATIONAL JOINT RESEARCH PROFECT EXPERIENCES

2020.06. – 2022.05	**Researcher**, Soundscape Attributes Translation Project (SATP) *PI: University College London, London, United Kingdom* *Co-I: Stockholm University (Sweden), Technical University of Berlin (Germany), McGill University (Canada), National University of Singapore (Singapore), etc.*
2018.07. – 2020.12.	**Researcher**, Science and Technology Amicable Relationships Program (STAR) *PI: Sorbonne University, Paris, France.*

REARCH INTERESTS

Main Topic	• **Spatial Audio & Immersive Sound Rendering (VR/AR, Auralization)** • **Perception-driven Audio Quality Evaluation (Psychoacoustics, EEG/HRV)** • **Soundscape-based Acoustic Design for Products & Urban Environments** • **EV / AVAS Sound Design and User-Centered Validation** • **Human-Centered Audio Modeling using AI & Data-driven Methods**
Details	Providing sustainable sound environment design guideline with human-centered models based on interdisciplinary approaches from acoustics, psychology, architecture, medical science, and urban planning with VR, ICT, deep learning.

HONORS, AWARDS AND FELLOWSHIPS

2025.10.	**Special Award for Research Paper** *Hyundai Motor Company (HMC)*
2022.11.	**Promising Scientist Award** *The Korean Society of Noise and Vibration Engineering (KSNVE)*
2022.03. – 2023.12.	**NST Postdoctoral Fellowship for Young Scientists** *Korea National Research Council of Science & Technology* (US$ 100,000)
2022.02.	**Ph.D. Excellence Paper Award**, *Hanyang University*.
2022.02.	**Best Paper Award**, *Hanyang University*.
2021.03.	**Superior Academic Performance** Excellence in Research Performance with papers, *Hanyang University*.
2020.09.	**LH President Award** Encouragement Award for LH National Idea Contest *Korea Land and Housing Corporation*.
2020.08.	**Young Professional Congress Award** *International Institute of Noise Control Engineering*, Inter-Noise 2020, Seoul, Korea. Paper Title: "Soundscape assessment methods: Compatibility of questionnaires and narrative interview based on ISO 12913-2", H. I. Jo, J. Y. Jeon (2020).
2020.06.	**NSV Scholarship** Grants for the encouragement of research *The Korean Society of Noise and Vibration Engineering (KSNVE)* (US$ 2,500)
2020.03.	**Superior Academic Performance Award** Excellence in Research Performance with papers, *Hanyang University*
2019.09.	**EAA Best Paper and Presentation Award** *European Acoustics Association (EAA)* International Congress on Acoustics (ICA) 2019, Aachen, Germany *Paper Title: "Soundscape cognition for user behavior in urban parks", H. I. Jo, J. Y. Jeon, J. Polack (2019)*
2019.09.	**Top 10 Best Poster Award** *Nederlands Akoestisch Genootschap* (NAG) The International Symposium on Room Acoustics (ISRA) 2019, Amsterdam, Netherlands *Poster Title: "Effects of stage volume ratio on acoustics of simulated concert halls" R. Seo, H. I. Jo, J. Y. Jeon (2019)*
2016.03. – 2020.08.	**BK 21 Plus Scholarship** Brain Korea 21 Plus Program, *Ministry of Education*, Korea (US$ 52,822)
2016.02.	**Graduation with Honors** Summa Cum Laude, *Hanyang University*
2015.03. – 2016.02.	**National Science & Technology Scholarship (Full tuition)** *Ministry of Science,* Korea (US$ 9,260)
2014.06.	**The 3rd prize** Technology management capstone design CEO camp, *Hanyang University*
2014.03. – 2015.12.	**Academic Excellence Scholarship (Full tuition)** *Hanyang University* (US$ 9,260)

PUBLICATIONS

h-index: 18, i10-index: 22, †: corresponding author (main author: 21 papers, Co-author: 3 papers)

Peer-Reviewed Journal Articles in Print or Accepted

1 **H. I. Jo**, J. Y. Jeon† (2024), "Quantification of visual attention by using eye-tracking technology for soundscape assessment through physiological response", *International Journal of Environmental Research and Public Health,* Vol. 21(11), pp. 1478.
Doi: https://doi.org/10.3390/ijerph21111478

2 F. Aletta†, A. Mitchell, T. Oberman, J. Kang, S. Khelil, T. A. K. Bouzir, D. Berkouk, H. Xie, Y. Zhang, R. Zhang, X. Yang, M. Li, K. Jambrošić, T. Zaninović, K. van den Bosch, T. Lühr, N. Orlik, D. Fitzpatrick, A. Sarampalis, P. Aumond, C. Lavandier, C. C. Moshona, S. Lepa, A. Fiebig, N. M. Papadakis, G. E. Stavroulakis, A. S. Sudarsono, S. J. Sarwono, G. E. Puglisi, F. Jafari, A. Astolfi, L. Shtrepi, K. Nagahata, **H. I. Jo**, J. Y. Jeon, B. Lam, J. Chieng, K. Ooi, J. Y. Hong, S. M. Antunes, S. Alves, M. L. de U. Carvalho, R. L. X. N. Michalski, P. Kogan, J. V. Manzano, R. G. Quesada, E. S. Silva, J. A. A. Pastor, M. E. Nilsson, Ö. Axelsson, W. S. Gan, K. N. Watcharasupat, S. Jaratjarungkiat, Z. T. Ong, P. N. D. Yörükoğlu, U. B. E. Osma, T. L. Nguyen, (2024). "Soundscape descriptors in eighteen languages: Translation and validation through listening experiments", *Applied Acoustics,* Vol. 224, pp. 110109.
Doi: https://doi.org/10.1016/j.apacoust.2024.110109
(Impact Factor: 3.4, Q1, Journal Ranking: 7/31 in Acoustics)

3 J. Y. Jeon, **H. I. Jo**†, K. S. Lee, (2023). "Psycho-physiological restoration with audio-visual interactions through virtual reality simulations of soundscape and landscape experiences in urban, waterfront, and green environments", *Sustainable Cites and Society,* Vol. 99, pp. 104929.
Doi: https://doi.org/10.1016/j.scs.2023.104929
(Impact Factor: 11.7, Q1, Journal Ranking: 1/68 in Construction & Building Technology)

4 **H. I. Jo**, K. S. Lee, J. Y. Jeon† (2022). "Effect of noise sensitivity on psychophysiological response through monoscopic 360 video and stereoscopic sound environment experience: a randomized control trial", *Scientific Report*, Vol. 12, pp. 1 – 12.
Doi: https://doi.org/10.1038/s41598-022-08374-y
(Impact Factor: 4.6, Q1, Journal Ranking: 21/133 in Multidisciplinary Sciences)

5 **H. I. Jo,** J. Y. Jeon† (2022). "Optimizing sound diffusion in a concert hall using scale-model measurements and simulations", *Journal of Building Engineering*, Vol. 50, pp.104240.
Doi: https://doi.org/10.1016/j.jobe.2022.104240
(Impact Factor: 6.4, Q1, Journal Ranking: 13/139 in Civil Engineering)

6 **H. I. Jo**, B. B. Santika, H. Lee, J. Y. Jeon† (2022). "Classification of sound environment based on subjective response with speech privacy in open plan offices", *Applied Acoustics*, Vol. 189, pp. 108595.
Doi: https://doi.org/10.1016/j.apacoust.2021.108595
(Impact Factor: 3.4, Q1, Journal Ranking: 7/31 in Acoustics)

7 **H. I. Jo**, B. B. Santika, H. Lee, J. Y. Jeon† (2022). "Influence of indoor soundscape perception based on audio–visual contents on work-related quality with preference and perceived productivity in open-plan offices", *Building and Environment,* Vol. 208, pp. 108598.
Doi: https://doi.org/10.1016/j.buildenv.2021.108598
(Impact Factor: 7.4, Q1, Journal Ranking: 6/139 in Civil Engineering and 7/68 in Construction and Building Technology)

8 J. Y. Jeon, **H. I. Jo†,** B. B. Santika, H. Lee (2022). "Crossed effects of audio-visual environment on indoor soundscape perception for pleasant open plan offices", *Building and Environment,* Vol. 207, pp. 108512.
Doi: https://doi.org/10.1016/j.buildenv.2021.108512

(Impact Factor: 7.4, Q1, Journal Ranking: 6/139 in Civil Engineering and 7/68 in Construction and Building Technology)

9 **H. I. Jo**, J. Y. Jeon† (2022). "Perception of urban soundscape and landscape using different visual environment reproduction methods in virtual reality", *Applied Acoustics*. Vol. 186, pp. 108498.
Doi: https://doi.org/10.1016/j.apacoust.2021.108498
(Impact Factor: 3.4, Q1, Journal Ranking: 7/31 in Acoustics)

10 J. D. Polack†, P. Taupin, **H. I. Jo**, J. Y. Jeon (2022). "Urban soundscapes in the imaginaries of native digital users: guidelines for soundscape design", *Sustainability*. Vol. 14(2), pp. 632.
Doi: https://doi.org/10.3390/su14020632
(Impact Factor: 3.9, Q2, Journal Ranking: 48/127 in Environmental Studies)

11 **H. I. Jo**, J. Y. Jeon† (2021). "Urban soundscape categorization based on individual recognition, perception, and assessment of sound environments", *Landscape and Urban Planning*, Vol. 216, pp. 104241.
Doi: https://doi.org/10.1016/j.landurbplan.2021.104241
(Impact Factor: 9.1, Journal Ranking: 1/43 in Urban Studies, 2/49 in Physical Geography, 2/86 in Geography, 7/169 in Ecology, and 3/40 in Regional & Urban Planning)

12 **H. I. Jo**, J. Y. Jeon† (2021). "Compatibility of quantitative and qualitative data-collection protocols for urban soundscape evaluation", *Sustainable Cities and Society,* Vol. 74, pp. 103259.
Doi: https://doi.org/10.1016/j.scs.2021.103259
(Impact Factor: 11.7, Q1, Journal Ranking: 1/68 in Construction & Building Technology)

13 **H. I. Jo,** J. Y. Jeon† (2021). "Overall environmental assessment in urban parks: Modelling audio-visual interaction with a structural equation model based on soundscape and landscape indices", *Building and Environment,* Vol. 204, pp. 108166.
Doi: https://doi.org/10.1016/j.buildenv.2021.108166
(Impact Factor: 7.4, Q1, Journal Ranking: 6/139 in Civil Engineering and 7/68 in Construction and Building Technology)

14 J. Y. Jeon, **H. I. Jo**†, K. S. Lee (2021). "Potential restorative effects of urban soundscapes: Personality traits, temperament, and perceptions of VR urban environments", *Landscape and Urban Planning,* Vol. 214, pp. 104188.
Doi: https://doi.org/10.1016/j.landurbplan.2021.104188
(Impact Factor: 9.1, Journal Ranking: 1/43 in Urban Studies, 2/49 in Physical Geography, 2/86 in Geography, 7/169 in Ecology, and 3/40 in Regional & Urban Planning)

15 Y. Chung, J. Jin, **H. I. Jo**, H. Lee, S. H. Kim†, S. J. Chung, H. J. Yoon, J. Park, J. Y. Jeon (2021). "Diagnosis of pneumonia by cough sounds analyzed with statistical features and AI". *Sensors,* Vol. 21(21), pp.7036
Doi: http://doi.org/10.3390/s21217036
(Impact Factor: 3.9, Q2, Journal Ranking: 19/63 in Instruments & Instrumentation)

16 **H. I. Jo**, J. Y. Jeon† (2020). "The influence of human behavioral characteristics on soundscape perception in urban parks: Subjective and observational approaches", *Landscape and Urban Planning,* Vol. 203, pp. 103890.
Doi: https://doi.org/10.1016/j.landurbplan.2020.103890
(Impact Factor: 9.1, Journal Ranking: 1/43 in Urban Studies, 2/49 in Physical Geography, 2/86 in Geography, 7/169 in Ecology, and 3/40 in Regional & Urban Planning)

17 **H. I. Jo**, J. Y. Jeon† (2020). "Effect of the appropriateness of sound environment on urban soundscape assessment", *Building and Environment*, Vol. 179, pp. 106975.
Doi: https://doi.org/10.1016/j.buildenv.2020.106975

(Impact Factor: 7.4, Q1, Journal Ranking: 6/139 in Civil Engineering and 7/68 in Construction and Building Technology)

18 J. Y. Jeon, **H. I. Jo**† (2020). "Effects of audio-visual interactions on soundscape and landscape perception and their influence on satisfaction with the urban environment". *Building and Environment*, Vol. 169, pp. 106544.
Doi: https://doi.org/10.1016/j.buildenv.2019.106544
(Impact Factor: 7.4, Q1, Journal Ranking: 6/139 in Civil Engineering and 7/68 in Construction and Building Technology)

19 J. Y. Jeon, **H. I. Jo**†, R. Seo, K. H. Kwak (2020). "Objective and subjective assessment of sound diffuseness in musical venues via computer simulations and a scale model". *Building and Environment*, Vol. 173, pp. 106740.
Doi: https://doi.org/10.1016/j.buildenv.2020.106740
(Impact Factor: 7.4, Q1, Journal Ranking: 6/139 in Civil Engineering and 7/68 in Construction and Building Technology)

20 J. Y. Jeon, R. Seo, **H. I. Jo**† (2020). "Effect of Stage Volume Ratio on Audience Acoustics in Concert Halls". *Sustainability*, Vol. 12(4), pp. 1370.
Doi: https://doi.org/10.3390/su12041370
(Impact Factor: 3.889, Q2, Journal Ranking: 57/127 in Environmental Studies)

21 J. Y. Jeon, **H. I. Jo**†, S. M. Kim, H. S. Yang (2019). "Subjective and objective evaluation of water-supply and drainage noises in apartment buildings by using a head-mounted display". *Applied Acoustics*, Vol. 148, pp. 289-299.
Doi: https://doi.org/10.1016/j.apacoust.2018.12.037
(Impact Factor: 3.4, Q1, Journal Ranking: 7/31 in Acoustics)

22 **H. I. Jo**, J. Y. Jeon† (2019). "Downstairs resident classification characteristics for upstairs walking vibration noise in an apartment building under virtual reality environment". *Building and Environment*, Vol. 150, pp. 21-32.
Doi: https://doi.org/10.1016/j.buildenv.2018.12.054
(Impact Factor: 7.4, Q1, Journal Ranking: 6/139 in Civil Engineering and 7/68 in Construction and Building Technology)

23 J. Y. Jeon, **H. I. Jo**† **(2019)**. "Three-dimensional virtual reality-based subjective evaluation of road traffic noise heard in urban high-rise residential buildings". *Building and Environment*, Vol. 148, pp.468-477.
Doi: https://doi.org/10.1016/j.buildenv.2018.11.004
(Impact Factor: 7.4, Q1, Journal Ranking: 6/139 in Civil Engineering and 7/68 in Construction and Building Technology)

24 J. Y. Jeon, H. S. Jang, **H. I. Jo**† **(2018)**. "Acoustic evaluation of orchestra occupancies in concert halls: Effect of sound absorption by orchestra members on audience acoustics". *Building and Environment*, Vol. 143, pp. 349-357.
Doi: https://doi.org/10.1016/j.buildenv.2018.07.028
(Impact Factor: 7.4, Q1, Journal Ranking: 6/139 in Civil Engineering and 7/68 in Construction and Building Technology)

Peer-Reviewed Journal Articles in Under Review or Prepared

1 Soundscape assessment of AVAS sounds of electrical vehicles in urban environments (Landscape and Urban Planning, 1st review)

2 A data-driven framework for detecting structural anomalies in construction procurement governance: A Case of nominal project bundling (Scientific report, 1st review)

Peer-Reviewed Papers in International Conference Proceedings

*: Presentation, **: Poster

1 **H. I. Jo**, H. Lee, B. B. Santika, J. Y. Jeon (2021). "Space dynamics for work performance enhancement in open plan office", *Euro-noise*, e-congress, Madeira, Portugal.*

2 **H. I. Jo**, J. Y. Jeon (2021). "Soundscape restoration model based on psycho-physiological response with audio-visual interaction in natural environment", *50th International Congress and Exposition on Noise Control Engineering (Inter-noise)*, e-congress, Wanshington DC, USA.*

3 **H. I. Jo**, J. Y. Jeon (2021). "Qualification of visual attention for static and dynamic environmental element with eye-tracking technique", *50th International Congress and Exposition on Noise Control Engineering (Inter-noise)*, e-congress, Wanshington DC, USA.*

4 B. B. Santika, **H. I. Jo**, H. Lee, J. Y. Jeon (2021) "Effect of visual elements on Indoor soundscape perception in open-plan office", *50th International Congress and Exposition on Noise Control Engineering (Inter-noise)*, e-congress, Wanshington DC, USA.*

5 H. Lee, B. B. Santika, **H. I. Jo**, J. Y. Jeon (2021). "Just noticeable difference of $L_{p,A,S,4m}$ specified ISO3382-3", *50th International Congress and Exposition on Noise Control Engineering (Inter-noise),* e-congress, Wanshington DC, USA.*

6 **H. I. Jo**, B. B. Santika, H. Lee, J. Y. Jeon (2021). "Just noticeable difference of ISO 3382-3 metrics for open-plan office noise", *13th ICBEN Congress on Noise as a Public Health Problem*, e-congress, Stockholm, Sweden.*

7 **H. I. Jo**, J. Y. Jeon (2021). "Overall environmental assessment with soundscape and landscape indices in urban parks", *13th ICBEN Congress on Noise as a Public Health Problem*, e-congress, Stockholm, Sweden.**

8 **H. I. Jo**, J. Y. Jeon (2020). "Compatibility of data collection protocol in ISO 12913-2 for urban soundscape assessment", *Forum Acusticum*, e-congress, Lyon, France.*

9 J. D. Polack, P. Taupin, **H. I. Jo**, J. Y. Jeon (2020). "Urban soundscapes in the imaginaries of digital native users in France and Korea", *Forum Acusticum*, e-congress, Lyon, France.*

10 F. Aletta, T. Oberman, O. Axelsson, H. Xie, Y. Zhang, S. Lau, S. Tang, K. Jambrošić, B. D. Coensel, K. V. Bosch, P. Aumond, C. Guastavino, C. Lavandier, A. Fiebig, B. Schulte-Fortkamp, J. Sarwono, A. Sudarsono, A. Astolfi, K. Nagahata, J. Y. Jeon, **H. I. Jo**, J. Chieng, W. S. Gan, J. Y. Hong, B. Lam, Z. T. Ong, P. Kogan, E. S. Silva, J. V. Manzano, P. N. Dökmeci Yörükoğlu, T. L. Nguyen, J. Kang (2020)." Soundscape assessment: towards a validated translation of perceptual attributes in different languages", *49th International Congress and Exposition on Noise Control Engineering (Inter-noise)*, Seoul, Korea.*

11 J. D. Polack, P. Taupin, **H. I. Jo**, J. Y. Jeon (2020.) "Guidelines for soundscape mapping and urban soundscape design derived from the imaginaries of digital native users", *49th International Congress and Exposition on Noise Control Engineering (Inter-noise)*, Seoul, Korea.*

12 **H. I. Jo**, R. Seo, J. Y. Jeon, (2020) "Soundscape assessment methods: Compatibility of questionnaires and narrative interview based on ISO 12913-2", *49th International Congress and Exposition on Noise Control Engineering (Inter-noise)*, Seoul, Korea.*
(Young Professional Award)

13 J. Y. Jeon, **H. I. Jo** (2020). "Soundscape assessment with classification of human perception to sound environment", *49th International Congress and Exposition on Noise Control Engineering (Inter-noise)*, Seoul, Korea.*

14 **H. I. Jo**, J. Y. Jeon, J. Polack (2019). "Soundscape cognition for user behavior in urban parks", *International Congress on Acoustics (ICA)*, Aachen, Germany.*
(EAA Best Paper and Presentation Award)

15 R. Seo, **H. I. Jo**, J. Y. Jeon (2019). "VR environment-based evaluation of impact factors on the urban soundscape recognition", *International Congress on Acoustics (ICA)*, Aachen, Germany.*

16 **H. I. Jo**, K. H. Kwak, J. Y. Jeon (2019). "Objective assessment of a scattered sound field in simulated concert halls and scale model", *The International Symposium on Room Acoustics (ISRA)*, Amsterdam, Netherlands.**

17 R. Seo, **H. I. Jo**, J. Y. Jeon (2019). "Effects of stage volume ratio on acoustics of simulated concert halls", *The International Symposium on Room Acoustics (ISRA)*, Amsterdam, Netherlands.**
(Top 10 Best Poster)

18 J. Y. Jeon, **H. I. Jo**, J. D. Polack (2019). "Evaluation of soundscape and audio-visual factors in VR environment for quiet urban park", *48th International Congress and Exposition on Noise Control Engineering (Inter-noise)*, Madrid, Spain.*

19 **H. I. Jo**, H. W. Kim, J. Y. Jeon (2019)." Investigation of work performance in open-plan office with soundscape variation using Virtual reality", *48th International Congress and Exposition on Noise Control Engineering (Inter-noise)*, Madrid, Spain.*

20 **H. I. Jo**, S. Ahmed, J. Y. Jeon (2018). "Annoyance evaluation of floor impact sounds with temporal and spatial variation in VR environments", *47th International Congress and Exposition on Noise Control Engineering (Inter-noise)*, Chicago, USA.*

21 S. Ahmed, **H. I. Jo**, J. Y. Jeon (2018). "Classification of human sounds using support vector machine with psychoacoustic data", *25th International Congress on Sound and Vibration (ICSV)*, Hiroshima, Japan.*

22 J. Y. Jeon, **H. I. Jo**, K. H. Kwak, H. W. Kim (2018). "The design goal of acoustic coefficients in concert halls related to reverberation and loudness", *Auditorium acoustics*, Hamburg, Germany.**

23 S. U. Park, **H. I. Jo**, M. Imran, J. Y. Jeon (2017). "A methodology for sound synthesis for graphical object interactions", *24th International Congress on Sound and Vibration (ICSV)*, London, UK.*

24 J. G. Seo, **H. I. Jo**, M. Imran, J. Y. Jeon (2017). "Evaluation of generalized cross correlation phased transform localization system in reverberation environments with a microphone array", *24th International Congress on Sound and Vibration (ICSV)*, London, UK.*

25 **H. I. Jo**, S. M. Kim, J. Y. Jeon, H. S. Yang (2017). "Measurement of water supply and drainage noise in multi-residential buildings", *46st International Congress and Exposition on Noise Control Engineering (Inter-noise)*, Hong Kong.*

Peer-Reviewed Published Abstract in International Conference

*: Presentation, **: Poster

1 **H. I. Jo,** J. Y. Hong, G. H. Kim, W. K. Moon (2024). Soundscape assessment for indoor acoustic environments in electric vehicle cabins, *22nd Treiennial Congress of the International Ergonomics Association*, Korea.*

2 J. Y. Jeon, **H. I. Jo**, K. S. Lee (2020). Potential restorative effect of the urban soundscape experience, *The Journal of the Acoustical Society of America (ASA)*, e-congress.*

3 **H. I. Jo**, J. Y. Jeon, K. S. Lee (2020). Restorative effect of interactive virtual reality natural environment experience on psychophysiological responses, *The Journal of the Acoustical Society of America (ASA)*, e-congress.*

4 J. Y. Jeon, **H. I. Jo**, J. D. Polack (2019). Comparing soundscape assessment method with natural language expression and questionnaire survey: A case study in Seoul, Korea, *178th The Journal of the Acoustical Society of America (ASA)*, San Diego, USA.*

5 J. Y. Jeon, **H. I. Jo** (2019). Topography for quiet areas and quite sides: real-time soundscape mapping for smart city, *Urban sound symposium*, Ghent, Belgium.*

6 **H. I. Jo**, H. S. Jang, J. Y. Jeon (2018). Sound diffusion design process using scale models of a concert hall and acoustic parameters, *176th The Journal of the Acoustical Society of America (ASA)*, Victoria, Canada.*

7 **H. I. Jo**, S. Ahmed, H. W. Kim, C. K. Seo, J. Y. Jeon (2018). VR-based realistic assessment of annoyance for the noise of children running in apartment buildings, *176th The Journal of the Acoustical Society of America (ASA)*, Victoria, Canada.*

8 **H. I. Jo**, J. G. Seo, J. Y. Jeon (2017). Detection of reflections in different room acoustical conditions using Generalized Cross Correlation Phase Transform, *174th The Journal of the Acoustical Society of America (ASA)*, New Orleans, USA.*

9 **H. I. Jo**, H. S. Jang, J. Y. Jeon (2017). Acoustical design of diffusers in concert halls using scale models, *The Journal of the Acoustical Society of America (ASA)*, Boston, USA.*

10 **H. I. Jo**, R. H. Afzal, I. Muhammad, J. Y. Jeon (2016). Sound synthesis methods of fire flame sound for predicting hidden images, *The Journal of the Acoustical Society of America (ASA)*, Salt lake, USA*

Peer-Reviewed Papers in Domestic Conference Proceedings

*: Presentation, **: Poster

1 **H. I. Jo**, J. Y. Jeon, J. Y. Hong (2025). "Perceptual and satisfaction models for AVAS based on soundscape design concept", *Hyundai Motor Group Conference*, Korea.*

Peer-Reviewed Published Abstract in Domestic Conference

*: Presentation, **: Poster

1 J. J. Yoon, J. Y. Hong, **H. I. Jo,** S. B. Kim (2025). "Evaluation of Electric Vehicle (EV) AVAS Based on Soundscape", *Conference of the acoustical society of Korea.*

J. J. Yoon, J. Y. Hong, **H. I. Jo,** K. H. Kim, W. K. Moon (2024). "AVAS soundscape evaluation according to electric vehicle (EV) driving conditions", *Conference of the acoustical society of Korea.*

2 **H. I. Jo,** (2022). "Soundscape design methods for pleasant indoor sound environment", *Conference of the Korean Society for Noise and Vibration Engineering.** ***(invited talk)***

3 **H. I. Jo,** J. Y. Lim, Y. M. Kim, S. E. Lee (2022). "The Influence of Occupant Behavior on Energy Consumption in Residential Buildings", *Conference of the Society of Air-conditioning and Refrigerating Engineers of Korea*.*

4 **H. I. Jo**, J. Y. Jeon (2021). "Quantification of visual attention for soundscape with eye-tracking technique", *Conference of the Korean Society for Noise and Vibration Engineering*.*

5 H. Lee, **H. I. Jo**, J. Y. Jeon (2021). "Emotional restoration effect of soundscape experience on dementia risk group", *Conference of the Korean Society for Noise and Vibration Engineering*.*

6 Y. Lee, **H. I. Jo**, J. Y. Jeon (2021). "Enhancement of cognitive function using non-invasive brain stimulation and 40 Hz sound stimulation", *Conference of the Korean Society for Noise and Vibration Engineering*.*

7 N. Y. Kim, **H. I. Jo**, J. Y. Jeon (2021). "Pneumonia diagnosis algorithm based on room acoustic variation", *Conference of the Korean Society for Noise and Vibration Engineering*.*

8 H. W. Yoon, **H. I. Jo**, J. Y. Jeon (2021). "Investigation of psycho-physiological response according to direction of heavy-weight floor Impact sound", *Conference of the Korean Society for Noise and Vibration Engineering*.*

9 A. Stepan, **H. I. Jo**, J. Y. Jeon (2021). "Effect of interaural cross-correlation function (IACF) on spatial impression of asymmetry binaural sound", *Conference of the Korean Society for Noise and Vibration Engineering*.*

10 J. S. Won, Y. Lee, **H. I. Jo**, K. S. Lee, J. Y. Jeon (2021). "Reduction of stress with mechanical massage using a massage-chair and music & imagery", *Conference of the Korean Society for Noise and Vibration Engineering*.*

11 Y. B. Chung, J. Jin, **H. I. Jo**, H. Lee, S. J. Chung, H. J. Yoon, J. Y. Jeon, S. H. Kim, J. S. Park (2020). "Determination of pneumonia symptoms through acoustic analysis of cough sound and AI-based algorithm", *Conference of the Korean Society for Noise and Vibration Engineering*.*

12 **H. I. Jo**, J. I. Kim, H. K. Park, D. H. Ahn, J. Y. Jeon (2020). "Psycho-physiological restorative effect with natural soundscape experience", *Conference of the Korean Society for Noise and Vibration Engineering*.*

13 K. J. Kim, **H. I. Jo**, G. S. Lee, S. H. Kim, J. Y. Jeon (2020). "Development of auditory hallucination assessment and psychoeducation system based on virtual reality for the patients with treatment resistant schizophrenia", *Conference of the Korean Society for Noise and Vibration Engineering*.*

14 **H. I. Jo**, D. H. Ahn, J. Y. Jeon (2019). "Soundscape analysis through comparative evaluation of brain wave and soundscape-quality protocol", *Conference of the Korean Society for Noise and Vibration Engineering*.*

15 J. Y. Park, **H. I. Jo**, H. Lee, S. H. Kim, J. Y. Jeon (2019). "Diagnosis of respiratory diseases using the bidirectional LSTM-attention model", *Conference of the Korean Society for Noise and Vibration Engineering*.*

16 R. Seo, J. H. Lee, **H. I. Jo**, H. K. Park, J. Y. Jeon (2019). "Investigation of acoustic condition inside incubator according to NICU sound environment", *Conference of the Korean Society for Noise and Vibration Engineering*.*

17 J. J. Jung, R. Seo, **H. I. Jo**, J. Y. Jeon (2019). "Correlation between acoustic parameters and the stage volume ratio of concert halls", *Conference of the Korean Society for Noise and Vibration Engineering*.*

18 **H. I. Jo**, Y. L. Choi, J. Y. Jeon (2019). "Evaluation of audio-visual factors for enhancement of urban soundscape under immersive virtual reality", *Conference of the Korean Society for Noise and Vibration Engineering*.*

19 H. W. Kim, **H. I. Jo**, J. Y. Jeon (2019). "Relation of task performance with Soundscape preference in open plan offices", *Conference of the Korean Society for Noise and Vibration Engineering*.*

20 H. W. Kim, **H. I. Jo**, R. Seo, W. Yang, J. Y. Jeon (2018). "A study on the correlation between masking effect and task performance by Soundscape in office space", *Conference of the Korean Society for Noise and Vibration Engineering*.*

21 **H. I. Jo**, R. Seo, J. Y. Jeon (2018). "Evaluation of preference for real-time urban soundscape", *Conference of the Korean Society for Noise and Vibration Engineering*.*

22 R. Seo, **H. I. Jo**, J. Y. Jeon (2018). "Acoustical design goal of concert halls for reverberation and loudness", *Conference of the Korean Society for Noise and Vibration Engineering*.*

23 H. W. Kim, **H. I. Jo**, J. Y. Jeon (2018). "Sound Insulation Performance of Wooden Frame Apartment", *Conference of the Korean Society for Noise and Vibration Engineering*.*

24 **H. I. Jo**, J. Y. Jeon (2018). "Correlation between urban soundscape variability and behavioral factors", *Conference of the Korean Society for Noise and Vibration Engineering*.*

25 J. I. Woo, **H. I. Jo**, J. Y. Jeon (2018). "Annoyance evaluation of water drainage noise in apartment building using HMD", *Conference of the Korean Society for Noise and Vibration Engineering*.*

26 S. Ahmed, J. I. Woo, **H. I. Jo**, J. Y. Jeon (2018). "Evaluation of Infiltrated Traffic Noise in Living Environment with VR Gear", *Conference of the Korean Society for Noise and Vibration Engineering*.*

27 S. M. Kim, **H. I. Jo**, S. U. Park, H. S. Yang, J. Y. Jeon (2017). "Analysis of plumbing noise characteristics based on perception tests", *Conference of the Korean Society for Noise and Vibration Engineering*.*

28 J. G. Seo, **H. I. Jo**, M. Imran, J. Y. Jeon (2017). "Evaluation of localization system based on linear microphone array with GCC-PHAT", *Conference of the Korean Society for Noise and Vibration Engin*eering.*

29 **H. I. Jo**, J. Y. Jeon (2017). "Acoustical design of diffusers in concert halls using scale models", *Conference of the Korean Society for Noise and Vibration Engine*ering.*

30 J. G. Seo, **H. I. Jo**, M. Imran, J. Y. Jeon (2016). "Evaluation of GCC-PHAT based real-time localization system in noisy environments", *Conference of the Korean Society for Noise and Vibration Engineering*.*

31 **H. I. Jo**, S. M. Kim, M. Imran, J. Y. Jeon (2016). "Perceptual evaluation of materials for virtual sounds reproduced by linear modal synthesis method", *Conference of the Korean Society for Noise and Vibration Engineering*.*

32 J. Y. Hong, **H. I. Jo**, J. Y. Jeon, (2016). "Soundscape evaluation and design methods in outdoor space of multi-dwelling houses", *Conference of the Korean Society for Noise and Vibration Engineering*.*

33 H. S. Jang, **H. I. Jo**, J. Y. Jeon (2016). "Evaluation of road traffic noise reduction by a tree scale model", *Conference of the Architectural Institute of Korea*.*

34 **H. I. Jo**, M. Imran, J. Y. Jeon (2016). "A virtual sound synthesis method for realization of fire sounds", *Conference of the acoustical society of Korea*.**

35 **H. I. Jo**, S. M. Kim, M. Imran, J. Y. Jeon (2016). "Evaluation of a stereophonic sound generation system using a 4-channel loudspeaker", *Conference of the acoustical society of Korea*.*

36 **H. I. Jo**, G. K. Seo, M. Imran, J. Y. Jeon (2016). "Real-time multiple sound location tracking system evaluation through MUSIC algorithm", *Conference of the acoustical society of Korea*.*

37 **H. I. Jo**, J. G. Seo, M. Imran, J. Y. Jeon (2016). "Evaluation of GCC-PHAT sound localization performance based on multi-channel microphone array", *Conference of the acoustical society of Korea*.*

Non-referred Invited Articles

– –

Published Reports

– –

INVITED SEMINARS AND PRESENTATIONS

Keynote and Plenary Lecture

1 Keynote: Sarah Payne, **H. I. Jo**, Daniel Steele (2021). "Public space soundscapes - three continents, three approaches", Urban Sound Symposium, E-congress, April, 2021.

Invited Seminar Speaker

1 **H. I. Jo** (2022). "Indoor Soundscape Design Based on Audio-Visual Interaction for Pleasant Open-Plan Offices", IJERPH Webinar | Promoting Healthy and Supportive Indoor Soundscapes, September, 2022.

Joint seminar

1 **H. I. Jo**, Soundscape Attributes Translation Project, September, 2019.
Managing department: University College London (UK)

2 **H. I. Jo**, Investigating the influence of space morphologic and behavioral factors on urban soundscape, Korea-France Joint Seminar (Hanyang University – Sorbonne University), November, 2018.

INTELLECTUAL PROPERTY

Patents

1 J. Y. Jeon, S. H. Kim, H. Lee, **H. I. Jo**, H. P. Zhao, H. M. Kim (2024). "Device for classifying sound source using deep learning, and method therefor", US-Patent No. 18273592

2 J. Y. Jeon, K. S. Lee, **H. I. Jo**, Y. N. Yun (2021). "Development of software for detecting eye tracking results in VR environment", Korea-Application No. 10-2021-0011413

3 J. Y. Jeon, S. H. Kim, H. Lee, **H. I. Jo**, H. P. Zhao, H. M. Kim (2021). "Improvement of Deep Learning based Sound Classifier Performance with Data Augmentation with Room Impulse Response", Korea-Application No. 10-2021-0011413

4 J. Y. Jeon, J. H. Park, K. S. Lee, **H. I. Jo**, K. J. Kim, H. J. Kim, J. S. Won, I. Yoon (2020). "Development of Extrapeutic Side Effects Evaluation System by Quantified Drugs", Korea-Application No. 10-2020-0177024

5 J. Y. Jeon, K. S. Lee, Y. Yoon, **H. I. Jo**, K. J. Kim, J. S. Won (2020). "Diagnostic and treatment software through the reproduction of virtual reality based hallucinations", Korea – Application No. 10-2020-0133015

6 J. Y. Jeon, M. Imran, J. G. Seo, B. J. Ryu, **H. I. Jo** (2018). "3D Sound Capturing Microphone Array for Wearable Displays and Mobile Devices", Korea - Registration No. 10-1678305

Technology Transfer (TT)

1 **Transfer Company: Hanyang S&A (US 50,000)** "Improvement of Deep Learning based Sound Classifier Performance with Data Augmentation with Room Impulse Response", Korea-Application No. 10-2021-0011413

INDUSTRY RESEARCH AND RESEARCH PROJECTS

Hyundai Motor Company

Project Title	**Vehicle interior sound environment design using soundscape concepts**
Period	Mar. 2023 – Present
Role	Senior Researcher (PI)
Detail	Industry-academia joint research (Chungnam National University) • *Hyundai/Kia/Genesis AVAS brand sound 2.0 design (EV3, IONIQ 5 AVAS tuning participation)* • *Competitor brand electric vehicle AVAS sound consumer evaluation (VR auralization with Ambisonic based multi-channel reproduction system)* • *Proposal new soundscape scale factors for electric vehicles sound* • *Proposal of sound environment quantification methodology (TDP) in car cabins applying architectural acoustic theory* Participation in company-wide AVAS expert committee activities

Project Title	**Vehicle VNH Test Research (K3, EV3, IONIQ5)**
Period	Aug. 2022 – Present
Role	Senior Researcher (PI)
Detail	B-Seg vehicle sedan/hatchback/high performance/PBV vehicle NVH test performance development • *New car performance development inspection and mass production schedule management* • *Certification of external noise in general area* • *Achieving the development goal of the starting master car stage* • *Preceding development and TCar development (Modified platform system performance target review, TCar development work)* Development of road noise element technology • *Analysis of big data correlation between load input point and indoor noise of small EV models* Knowledge structure • *A study on ways to reduce the idling vibration contribution of bumpers/hoods*

FUNDED RESEARCH AND RESEARCH PROJECTS

Extramural Competitive Research Grants

Project Title	**Data-Centric Checkup Technique of Building Energy Performance**
Funding Source	*Ministry of Science and ICT*
Period	Mar. 2022 – Dec. 2022
Grant Amount	US 1,705,000
Role	Researcher

Project Title	**2022 Zero Energy Building Support Center Operation Project**
Funding Source	*Ministry of Science and ICT*
Period	Mar. 2022 – Dec. 2022
Grant Amount	US 400,000
Role	Researcher

Project Title	**Development of Digital Therapeutics Using Social Recognition-based VR Soundscape for Mental Illness**
Funding Source	*National Research Foundation of Korea*
Period	Mar. 2020 – Feb. 2022
Grant Amount	US 600,000
Role	Research Assistant

Project Title	**Development of medical solutions with vibro-acoustics**
Funding Source	*National Research Foundation of Korea*
Period	Jul. 2019 – Feb. 2022
Grant Amount	US 1,400,000
Role	Research Assistant

Project Title	**Office product and space planning guide through sound environment analysis according to product and space conditions**
Funding Source	*FURSYS Inc.*
Period	Dec. 2019 – Apr. 2020
Grant Amount	US 60,000
Role	Research Assistant

Project Title	**Investigation of the localization of individual instruments inside the orchestra and the acoustical design of halls for Korean traditional music**
Funding Source	*National Research Foundation of Korea*
Period	Nov. 2018 – Oct. 2020
Grant Amount	US 1,500,000
Role	Research Assistant

Project Title	**Investigation the morphological and behavior factors in urban soundscape**
Funding Source	*National Research Foundation of Korea*
Period	Jul. 2018 – Dec. 2020
Grant Amount	US 30,000
Role	Research Assistant

Project Title	**A study on spatial acoustic-based risk situation analysis modeling and performance evaluation**
Funding Source	*Electronics and telecommunications research institute (ETRI)*
Period	May. 2018 – Oct. 2018
Grant Amount	US 30,000
Role	Research Assistant

Project Title	**Research on sound walking based on spatial sound technology (capturing / noise cancellation / risk analysis)**
Funding Source	*Electronics and telecommunications research institute (ETRI)*
Period	May. 2017 – Oct. 2017
Grant Amount	US 30,000
Role	Research Assistant

Project Title	**A Creative Expertise Development Team for Integrated Building Performance Enhancement**
Funding Source	*National Research Foundation of Korea*
Period	Mar. 2016 – Aug. 2020
Grant Amount	US 878,188
Role	Research Assistant

Project Title	**Development of Real-time Modeling and Rendering Methodology of Dynamic Interactive 3D Sound in Coexistent Space**
Funding Source	*National Research Foundation of Korea*
Period	Jan. 2016 – Aug. 2019
Grant Amount	US 740,000
Role	Research Assistant

Project Title	**Auditory evaluation and questionnaire for water supply and drainage noise**
Funding Source	*Korea Land and Housing Corporation*
Period	Sep. 2016 – Apr. 2017
Grant Amount	US 60,000
Role	Research Assistant

Intramural Competitive Research Grants

Project Title	**Vibro-acoustics medical lab**
Funding Source	*Hanyang University*
Period	May. 2019 – Apr. 2021
Grant Amount	US 30,000
Role	Research Assistant

Concert Hall Design Projects

Project Title	**Acoustical design of concert hall of Bucheon Culture Arts Center**
Use	*Multi-purpose theater*
Period	Mar. 2018 – Feb. 2022
Seats	1,450
Role	Senior Researcher

Project Title	**Acoustical design of concert hall and opera house of IFEZ Art Center**
Use	*Multi-purpose theater*
Period	Dec. 2010 – Nov. 2016
Seats	1,800
Role	Research Assistant

PROFESSIONAL CREDENTIAL

2020.07. – Present	**Advanced Data Analytics Semi-Professional (ADsP)** *Korea Data Agency,* National certification from Korea government
2017.01. – Present	**WELL AP** *International WELL Building Institute (IWBI),* Demonstrating the knowledge and understanding of the interaction between health and wellness within the built environment and the implementation process for the WELL Building Standard™
2017.02. – Present	**LEED AP® Interior Design + Construction (ID+C)** *U.S. Green Building Council,* Demonstrating the knowledge and understanding of green building practices and principles needed to support the use of the LEED green building program.
2016.12. – Present	**LEED AP® Building Design + Construction (BD+C)** *U.S. Green Building Council,* Demonstrating the knowledge and understanding of green building practices and principles needed to support the use of the LEED green building program.

SKILLS AND TECHNIQUES

2018.01.	Acoustic basic theory (25th noise control) education, KAIST
2017.12.	Completion of PULSE User training (General), BRÜEL & KJÆR
2017.11.	Completion of noise vibration analysis training, BRÜEL & KJÆR
2017.10.	Completion of the training for Coding for Deep Learning, KSNVE
2016.07.	Completion of MATLAB for Data Processing and Visualization, MathWorks
2016.07.	Completion of MATLAB Fundamentals training, MathWorks
Language	MATLAB, R Studio, Python
Statistics	SPSS, R Studio, Jamovi, Gephi
Software	Test lab, INCA, Odeon, Labshop, Pulse, 01dB, AS-70, AMFG, ArcGIS, Unity3D, Auto CAD, Sketchup
Hardware	Microphone (1/2, 1/4, 1/8 inch, ambisonic, spherical, etc.), Accelerometer, Recorder, Multi-channel speaker, Head-mounted display, Motion tracker, etc.

SERVICE EXPERIENCE

COMMITTEES

Professional

–	–

EDITORSHIP AND REVIEWER ACTIVITIES

Editorship

–	–

Journal Reviewer

2023 – Present	Environmental Research, Elsevier
2023 – Present	Sustainable Cities and Society, Elsevier
2023 – Present	Journal of Building Engineering, Elsevier
2021 – Present	Landscape and Urban Planning, Elsevier
2021 – Present	Building and Environment, Elsevier
2021 – Present	Science of the Total Environment, Elsevier
2021 – Present	Ecological Indicators, Elsevier
2021 – Present	International Journal of Environmental Research and Public Health, MDPI
2021 – Present	Applied Acoustics, Elsevier
2021 – Present	Applied Science, MDPI
2021 – Present	Frontiers in Psychology, Frontiers Media
2021 – Present	Noise Mapping, De Gruyter Poland sp zoo
2021 – Present	Buildings, MDPI
2021 – Present	IEEE Access, IEEE
2021 – Present	Journal of Environmental Planning and Management, Routledge Journals
2021 – Present	Computer Modeling in Engineering & Sciences, Tech science press
2021 – Present	AIMS Environmental Science, AIMS
2021 – Present	Journal of Asian Architecture and Building Engineering, Taylor & Francis LTD.

CONFERENCE MODERATOR AND FACILITATOR

–	–

AFFILIATIONS

Member	The Acoustical Society of America (ASA)
Member	The Acoustical Engineering Society (AES)
Member	The Korean Society for Noise and Vibration Engineering (KSNVE)
Member	The Acoustical Society of Korea (ASK)
Member	The Architectural Institute of Korea (AIK)
Member	The Society of Air-Cooling and Refrigerating Engineering of Korea (SAREK)

TEACHING EXPERIENCE

COURSES TAUGHT

Teaching at Hanyang University (HYU)

–	–

CURRENT AND FORMER GRADUATE STUDENTS

–	–

CURRENT AND FORMER POSTDOCTORAL RESEARCHERS SUPERVISED

–	–

UNDERGRADUATE STUDENTS SUPERVISED IN INDEPENDENT RESEARCH

–	–

OTHER TEACHING ACCOMPLISHMENTS AND PROFESSIONAL DEVELOPMENT

2015.08	Completed Hanyang summer architecture camp for High school student Hanyang University, August 10th – 12th, 2015